

MCoT-MVS: Multi-level Vision Selection by Multi-modal Chain-of-Thought Reasoning for Composed Image Retrieval

Xuri Ge*
xuri.ge@sdu.edu.cn
Shandong University
Jinan, China

Chunhao Wang*
Shandong University
Qingdao, China
202415164@mail.sdu.edu.cn

Xindi Wang
Shandong University
Jinan, China
xindi.wang@sdu.edu.cn

Zheyun Qin
Shandong University
Qingdao, China
zheyunqin@gmail.com

Zhumin Chen
Shandong University
Jinan, China
chenzhumin@sdu.edu.cn

Xin Xin†
Shandong University
Qingdao, China
xinxin@sdu.edu.cn

Abstract

Composed Image Retrieval (CIR) aims to retrieve target images based on a reference image and modified texts. However, existing methods often struggle to extract the correct semantic cues from the reference image that best reflect the user's intent under textual modification prompts, resulting in interference from irrelevant visual noise. In this paper, we propose a novel Multi-level Vision Selection by Multi-modal Chain-of-Thought Reasoning (MCoT-MVS) for CIR, integrating attention-aware multi-level vision features guided by reasoning cues from a multi-modal large language model (MLLM). Specifically, we leverage an MLLM to perform chain-of-thought reasoning on the multimodal composed input, generating the retained, removed, and target-inferred texts. These textual cues subsequently guide two reference visual attention selection modules to selectively extract discriminative patch-level and instance-level semantics from the reference image. Finally, to effectively fuse these multi-granular visual cues with the modified text and the imagined target description, we design a weighted hierarchical combination module to align the composed query with target images in a unified embedding space. Extensive experiments on two CIR benchmarks, namely CIRRR and FashionIQ, demonstrate that our approach consistently outperforms existing methods and achieves new state-of-the-art performance. Code and trained models are publicly released at <https://github.com/JJJJerry/WWW2026-MCoT-MVS>.

CCS Concepts

• **Information systems** → **Query reformulation; Novelty in information retrieval; Combination, fusion and federated search.**

Keywords

Composed Image Retrieval, Multi-modal Chain-of-Thought Reasoning, Vision Selection

*Both authors contributed equally to this research.

†Corresponding author



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1 Introduction

Composed Image Retrieval (CIR) [26] is a fundamental task in multimodal understanding, aiming to retrieve the target image corresponding to the user's intention based on the given reference image and modified texts. It requires a thorough understanding of the modification intent implicit in the modified texts to accurately perceive the relevant contents in the reference image. CIR task has garnered increasing attention in both academia and industry due to its broad real-world applications, such as personalized recommendations in e-commerce [7, 32] and interactive search engines [17, 33], etc. The challenge of the CIR task is to correctly understand the user's intent from multimodal combined queries. In particular, it is more challenging to integrate and align visual information from diverse reference image semantics related to the modification intent of the modification text to construct a joint representation that matches the target image.

Early CIR methods [8, 9, 13] primarily rely on joint embedding space modeling by fusing the global-level representations of the reference image and the modification text to match candidate target images. As shown in Figure 1(B), these methods typically employ direct or low-level fusion strategies across modalities, lacking fine-grained semantic alignment. As a result, they struggle to accurately capture the visual modification contents based on the modified text. Especially when it comes to attribute detail changes and instance-level semantic replacement, the crude visual perception ability leads to visual noise in the combined query features.

To improve the representation ability of multimodal composed queries, recent methods [4, 24, 29] incorporate cross-modal attention mechanisms to establish local or global semantic associations between reference images and modification texts, as illustrated in Figure 1(C). Attention-based fusion can enhance both the intent-aware representation of composed queries and their semantic consistency with potential target images. For example, Xu et al. [34] introduces multiscale attention-aware correspondences between

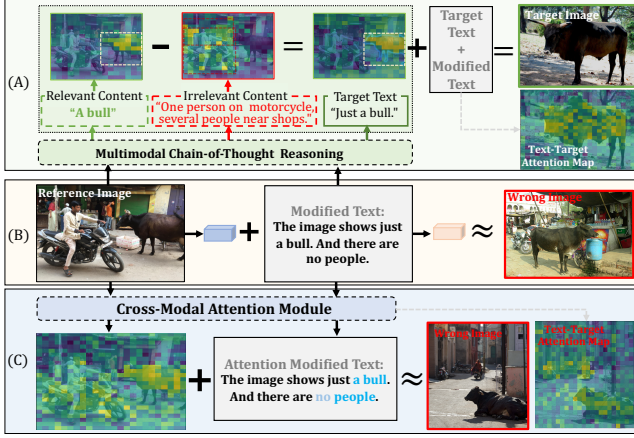


Figure 1: Compared with the traditional CIR model (B) and cross-modal attention-based CIR model (C), in this paper, we introduce multimodal chain-of-thought reasoning to explicitly extract the relevant and irrelevant contents for reference visual information selection (A). In this way, while effectively focusing on fine-grained useful references, we can significantly avoid the attention noise interference driven by visual-linguistic correlation as in method (C).

reference images and modified texts to enhance composed-query representations, while Tian et al. [24] employs additive attention to capture the intended semantics from multimodal queries. Despite the certain effectiveness of attention mechanisms in improving joint multimodal representations, existing methods face clear limitations in interpreting complex composed-query intents and selecting visual regions aligned with them. Specifically, these methods typically rely on shallow appearance semantic correlations between salient objects in the reference image and the modification text, lacking explicit semantic reasoning. As shown in Figure 1(C), they struggle to distinguish implicit visual operations in the modification, such as retaining “a bull” or removing “all people, motorcycle, and shop”, etc. This results in reference image content (e.g., “a people on a motorcycle”), which should be removed, being incorrectly included in the combined query due to a spurious semantic similarity attention mechanism. This undermines the retrieval intent and degrades retrieval performance.

To deal with the above issue, we argue that an intuitive motivation shown in Figure 1(A) is to decompose the user’s composed query intent and integrate the correct reference visual semantics aligned with the intended modification, thereby reducing the ambiguity of the composed query. On the one hand, the rapid progress of Large Language Models (LLMs) offers powerful capabilities for intent reasoning. The latest zero-shot CIR methods [21, 22] also introduce to reason about user modification intentions based on LLMs. However, these studies primarily rely on textual cues without effective interaction with the visual modality, resulting in composed queries still being polluted by irrelevant visual features. On the other hand, effectively distinguishing the useful reference image elements is crucial for enhancing visual grounding in composed queries. The representative solutions are in two folds. One is to process the global grid features (i.e., patch) [13, 28], and the other

is instance-level representations based on object detection [10]. While attention-aware visual interactions are then performed separately on these features, their selection is often still biased by visual-linguistic similarity, lacking explicit intent grounding.

Therefore, integrating intent decomposition with effective reference visual feature selection is a natural choice for explicitly modeling composed-query intent and enhancing multimodal representations. As manifested in Figure 1(A), reasoning explicitly about both relevant and irrelevant elements, along with the target context, enables more precise localization of visual intention and avoids distraction from irrelevant content, compared to Figure 1(C).

In this paper, we propose a novel CIR framework, **MCot-MVS** (Multi-level Vision Selection via Multi-modal Chain-of-Thought Reasoning). The framework first performs chained reasoning on multimodal composed queries using a multimodal large-scale language model (MLLM) to generate structured linguistic cues that explicitly mark semantic content to be retained, removed, and user-intended goals. Subsequently, we design two vision selection modules to operate on both patch-level and instance-level features of the reference image, enabling multi-granularity extraction of semantically aligned visual cues. Finally, a weighted hierarchical combination (WHC) module fuses the selected visual features with the modification and referred target text to further clarify user intent, aligning the final composed representation with the correct target image in a shared embedding space.

The contributions of this paper are as follows:

- We propose a novel MCot-MVS with a multi-modal chain-of-thought reasoning for explicit multi-level reference visual element selection with hierarchical combination. It effectively reduces irrelevant multimodal noise of queries in composed image retrieval.
- We design multi-level reference vision selection modules, including a Patch-level Visual Reference Selection (PVRS) and an Instance-level Visual Reference Selection (IVRS), which decompose and select reference image contents at both fine-grained and high-level semantics. This improves the accuracy of the reference visual intent.
- We introduce a Weighted Hierarchical Combination (WHC) to effectively fuse selected visual cues with modified text and reasoned user-intended text in a unified embedding space, enabling precise alignment with target images.

The proposed MCot-MVS is verified to achieve the state-of-the-art (SOTA) retrieval performance on two standard composed image retrieval benchmarks, *i.e.* CIRR and FashionIQ.

2 Related Works

CIR with Direct Composed Query Fusion. Early approaches to Composed Image Retrieval (CIR) [3, 13, 28] focus on directly combining the reference image and the modified text to construct a joint query representation. These methods typically adopt either global feature concatenation or local feature fusion. For example, Chen and Bazzani [3] proposed a unified joint visual semantic matching model that learns image-text directly compositional embeddings by jointly associating visual and textual modalities in a shared discriminative embedding space via compositional losses. Kim et al. [13] introduced a dual composition network to learning a joint

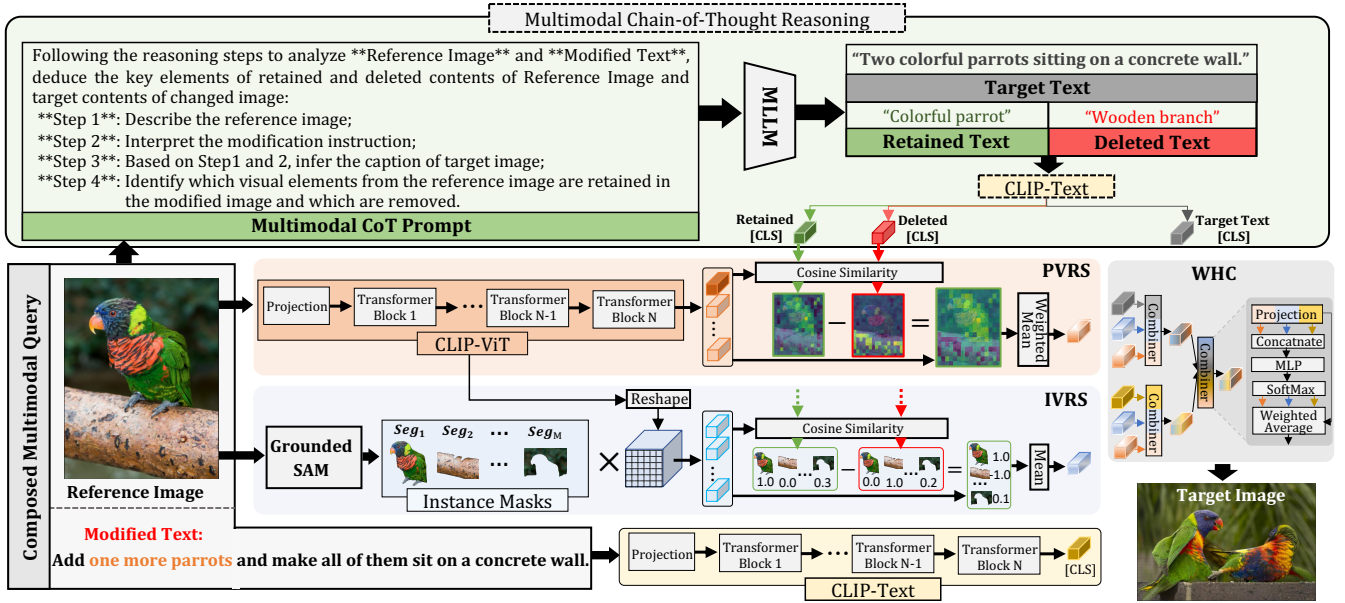


Figure 2: Illustration of our proposed MCoT-MVS. It primarily consists of the Multimodal Chain-of-Thought (CoT) Reasoning module, Patch-level Visual Reference Selection (PVRs), the Instance-level Visual Reference Selection module (IVRS), and a Weighted Hierarchical Combination (WHC). The multimodal CoT reasoning module first reasons the composed query into retained and deleted contents, as well as the potential target context, based on a pre-trained MLLM. Then, the useful patch-level and instance-level reference visual representations are semantically selected by the explicitly reasoned retained and deleted texts to retain the correct user intent and to remove visual noise. Finally, the WHC fuses the selected representations with multiple target texts into an attention-aware query, aligning with the target image.

embedding space by constructing two cyclic directional mappings for global embedding of triplets (reference image, text query, target image) via both composition and correction. While these methods offer a straightforward way to integrate modalities, they often struggle to disentangle which aspects of the composed query should be retained, removed, or replaced based on the modification intent.

CIR with Attention-based Composed Query Fusion. To improve the semantic alignment between reference images and modified text, attention-based CIR models [6, 24, 29] have been developed. These approaches typically learn cross-modal interactions via self- and cross-attention modules between the multimodal composed queries. For instance, Delmas et al. [6] ensembled multiple target-query similarity scores by exploring the attention mechanisms between the reference image and modified text, as well as target images and modified text. [2] proposed a simple soft attention-aware combiner module to compose the multimodal queries from the pre-trained foundation vision-language models, achieving the effective fusion of multimodal combination features from the pre-trained model. However, they usually rely on shallow similarity patterns and lack deeper semantic reasoning capabilities, which limits their ability to understand complex modification intents.

CIR with LLM-guided Intent Reasoning and Understanding. Recent advances leverage Large Language Models (LLMs) and Multi-modal Large Language Models (MLLMs) to enhance CIR with high-level reasoning. Some works [12] introduce intent rewriting, where the composed query is reformulated into a target description using LLMs. It has shown strong generalization in zero-shot CIR settings. Others, such as CIR-LVLM [21, 22], fine-tuned VLM

models to align image-text features through latent intent inference. These methods benefit from external knowledge and zero-shot generalization but typically treat the reference image as a whole, without fine-grained control over different levels of visual content. Our work differs by explicitly modeling multi-level vision selection (patch-level and instance-level) guided by MLLM-based multi-step reasoning, enabling interpretable and controllable fusion of reference semantics.

3 The Proposed Method – MCoT-MVS

The overview of MCoT-MVS is illustrated in Figure 2. The innovation of MCoT-MVS lies in improving the consistency between multimodal composed queries and target images by effectively decomposing composed query intent (Multimodal CoT Reasoning), selecting useful multi-level query references (PVRs and IVRS) and reconstructing multimodal combined query representation (WHC).

3.1 Preliminary

We propose a novel MCoT-MVS approach for CIR, aiming to effectively retrieve the target image I^T given a multimodal composed query consisting of a reference image I^R and the corresponding modification texts T . During training stage, given the triplet training set $\mathcal{G} = \{(I_i^R, T_i, I_i^T)\}_{i=1}^{\Omega}$, a joint query-target embedding space is optimized by narrowing the embedding distance between the representation of the multimodal query ($\mathcal{F}(I_i^R, T_i)$) and the corresponding representation of target image ($\mathcal{H}(I_i^T)$). In this paper, we employ a widely used loss function in the CIR research community,

i.e., batch-based classification loss [26], formulated as,

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^B -\log \left(\frac{\text{Exp}\{\text{Cos}(\mathcal{F}(I_i^R, T_i), \mathcal{H}(I_i^Q))/\tau\}}{\sum_{j=1}^B \text{Exp}\{\text{Cos}(\mathcal{F}(I_i^R, T_i), \mathcal{H}(I_j^Q))/\tau\}} \right), \quad (1)$$

where B is the batch size, τ is the temperature coefficient and $\text{Cos}(\cdot, \cdot)$ denotes the cosine similarity function. Our final aim is to optimise all parameters of $\mathcal{F}$ and $\mathcal{H}$, where $\mathcal{F}()$ for the composed query encoding contains the main contributions of our proposed MCoT-MVS shown in Figure 2. Algorithm 1 details the training procedure of MCoT-MVS.

3.2 Multimodal Chain-of-Thought Reasoning

To fully understand the user's composed query intention, especially the heterogeneous and complex expression of multimodal queries, we first propose a multimodal chain-of-thought (CoT) reasoning method based on a pre-trained MLLM to effectively decompose the multimodal query intention and explicitly separate the query intention. Similar to [21], as shown in Figure 2, we design a multi-step CoT reasoning prompt for the pretrained MLLM, including reference image understanding, modified text understanding, content inference of the corresponding target image, and decomposition of the visual elements to be retained or deleted. Different to other CoT reasoning CIR models [21], we use a unified multimodal CoT reasoning prompt to decompose and reorganise the query intentions, which maintains both high efficiency and effectiveness. After that, the multimodal composed query is decomposed into the retained text RT and the deleted text DT and is reconstructed into a potential target text TT .

$$RT, DT, TT = \text{MLLM}(I^R, T; \text{CoT Prompt}), \quad (2)$$

Then, we employ the text encoder of CLIP [18] to extract the representations of these reasoning results as follows:

$$\{R^{RT}, R^{DT}, R^{TT}\} = \text{CLIP-Text}(RT/DT/TT), \quad (3)$$

where $\{R^{RT}, R^{DT}\}$ will be used for the multi-level visual reference selection and R^{TT} for final query combination.

3.3 Multi-level Visual Reference Selection

Unlike the explicit semantics of the modified text, selecting relevant visual elements from the reference image in CIR is more challenging, requiring both accurate intent understanding and effective filtering of irrelevant content to reduce ambiguity. To enhance the effectiveness of reference image representation, we propose patch-level and instance-level visual reference selection modules, termed PVRS and IVRS. Guided by the preceding reasoning results, i.e. R^{RT} for retained semantic and R^{DT} for deleted semantic, we explicitly reconstruct attention for both patch-level and instance-level visual reference features based on inferred user intent. This mechanism emphasizes the retained visual elements while suppressing those related to significant modifications or replacements, thereby improving the reference image's relevance to the composed query.

For the fine-grained PVRS module, we first extract the patch-level representation of the reference image I^R using the CLIP-ViT encoder [18], denoted as $V^P = \{[CLS], v_1^P, \dots, v_N^P\}$, where v_i^P is the feature of i -th visual patch. Then, we leverage the previously generated reasoning representations R^{RT} and R^{DT} to compute the cosine

Algorithm 1: Pseudo code of MCoT-MVS.

Input: Training set $\mathcal{G} = \{(I_i^R, T_i, I_i^T)\}_{i=1}^{\Omega}$, pre-trained MLLM, pre-trained Grounded-SAM, pre-trained CLIP, batch size B , epochs E

Output: Trained MCoT-MVS model

```

1 Initialize CLIP-ViT, CLIP-Text, PVRS, IVRS, WHC modules;
2 for epoch = 1 to E do
3   Shuffle  $\mathcal{G}$ ; split into batches of size  $B$ ;
4   for each batch do
5     for each composed query  $(I_i^R, T_i)$  do
6       // Step 1: Multi-modal CoT Reasoning
7        $\{RT_i, DT_i, TT_i\} \leftarrow \text{MLLM}(I_i^R, T_i, \text{CoTPrompt})$ ;
8        $\{R_i^{RT}, R_i^{DT}, R_i^{TT}\} \leftarrow \text{CLIP-Text}(RT_i/DT_i/TT_i)$ ;
9       // Step 2: Multi-level Visual Reference Selection
10       $\hat{V}_i^P \leftarrow \text{PVRS}(\text{Eq.}(4-6))$ ;
11       $\hat{V}_i^I \leftarrow \text{IVRS}(\text{Eq.}(7-9))$ ;
12      // Step 3: Weighted Hierarchical Combination
13      Encode modification text:  $S_i^{MT} = \text{CLIP-Text}(T_i)$ ;
14       $Q_i^M = \text{Combiner}(\hat{V}_i^P, \hat{V}_i^I, S_i^{MT})$ ;
15       $Q_i^T = \text{Combiner}(\hat{V}_i^P, \hat{V}_i^I, R_i^{TT})$ ;
16       $\mathcal{F}_i = \text{Combiner}(Q_i^M, Q_i^T)$ ;
17    end
18    Extract target image:  $\mathcal{H}_i = \text{CLIP-ViT}(I_i^T)$ ;
19    Update parameter of CLIPs, PVRS, IVRS, WHC
20    modules according to Loss function  $\mathcal{L}$  in Eq.1.;
21  end

```

similarity with each visual patch, serving as attention weights for identifying retained and deleted areas.

$$\alpha_i^{P+} = \text{Cos}(v_i^P, R^{RT}) = \frac{v_i^P \cdot R^{RT}}{\|v_i^P\| \|R^{RT}\|}, \quad (4)$$

$$\alpha_i^{P-} = \text{Cos}(v_i^P, R^{DT}) = \frac{v_i^P \cdot R^{DT}}{\|v_i^P\| \|R^{DT}\|}, \quad (5)$$

The final refined patch-level reference visual representation $\hat{V}^P$ is obtained by contrastively re-weighting the original feature with the retain/delete attentions, formulated as,

$$\hat{V}^P = \left(\frac{1}{N} \sum_{i=1}^N (\alpha_i^{P+} - \alpha_i^{P-}) v_i^P + V_{[CLS]}^P \right) / 2, \quad (6)$$

where N is the number of patches of the reference image. This emphasises reference visual patches aligned with retained semantics and suppresses visual noise, enhancing the fine-grained semantic relevance in CIR.

Beyond patch-level feature selection, we further propose a high-level IVRS module, which captures more complete instance semantics and shape-aware information. Specifically, we first utilize the pre-trained Grounded SAM model [19] to segment the reference image I^R into M potential instance areas, denoted as $\text{Seg}_{i=1}^M$. These instance masks are then used to extract the visual features of the reference objects $O^R = o_1^R, \dots, o_M^R$ via masked average pooling over

the CLIP-ViT encoder [18] feature maps. Subsequently, we compute the cosine similarity between each instance feature o_i^R and the previously generated reasoning representations for "Retain" and "Delete" semantics, R^{RT} and R^{DT} , to obtain the corresponding attention weights α_i^{RT} and α_i^{DT} :

$$\alpha_i^{I^+} = \frac{v_i^I \cdot R^{RT}}{\|v_i^I\| \|R^{RT}\|}, \alpha_i^{I^-} = \frac{v_i^I \cdot R^{DT}}{\|v_i^I\| \|R^{DT}\|}, \quad (7)$$

We then calculate the instance attention weights $\tilde{\alpha}_i^I$ by difference the retained weights and deleted weights after min-max normalization. Finally, the refined instance-level reference visual representation $\hat{V}^I$ is aggregated by weighted mean:

$$\tilde{\alpha}_i^{I^*} = \frac{\alpha_i^{I^*} - \min(\{\alpha^{I^*}\})}{\max(\{\alpha^{I^*}\}) - \min(\{\alpha^{I^*}\})}, \quad (8)$$

$$\hat{V}^I = \frac{1}{M} \sum_{i=1}^M (\tilde{\alpha}_i^{I^+} - \tilde{\alpha}_i^{I^-}) v_i^I, \quad (9)$$

where I^* is I_+ or I_- .

3.4 Weighted Hierarchical Combination

After multi-level visual selection guided by explicit reasoning cues from multimodal CoT reasoning, we obtain multiple multimodal query representation, including the patch-level visual reference feature $\hat{V}^P$, the instance visual reference feature $\hat{V}^I$, the modified textual feature S^{MT} extracted from the CLIP-Text encoder, and the reasoned target textual feature R^{TT} . To effectively integrate these multimodal representations and accurately reflect user intent, we propose a Weighted Hierarchical Combination (WHC) module. This module performs hierarchical and adaptive fusion of the different query features under alignment supervision with the target image.

Specifically, we first fuse the modified feature with multi-level visual features using adaptive weights, allowing features that more clearly reflect the user's intent to be emphasized. In parallel, we integrate the inferred target semantics with visual references to mitigate noise from incorrect reasoning. Inspired by [2], we revise the *Combiner* (see Figure 2) with triplet inputs (Corresponding textual representation, PVRs's visual representation and IVRS's visual representation). It consists of a concatenation operation with projecting, an MLP attention computation, and a weighted aggregation. We take the Modification *Combiner* process as an example, formulated as,

$$V^{cat} = [W_1 \hat{V}^P : W_2 \hat{V}^I : W_3 S^{MT}], \quad (10)$$

$$\{\beta_1, \beta_2, \beta_3\} = \text{SoftMax}(\text{MLP}(V^{cat})), \quad (11)$$

$$Q^M = \beta_1 * \hat{V}^P + \beta_2 * \hat{V}^I + \beta_3 * S^{MT} + \text{MLP}(V^{cat}). \quad (12)$$

A final *Combiner* fusion combines both streams Q^M and Q^T into a unified query representation Q .

Finally, the unified query representation Q from our proposed MCoT-MVS (i.e. $\mathcal{F}()$) is used to align with the target image feature (i.e. $\mathcal{H}(I^T)$).

Table 1: Performance comparison on CIRR from the open-world scene. The best and the second-best scores are marked in bold and underlined fonts, respectively.

Method	R@1	R@5	R@10	R@50	Avg
<i>Traditional Model-Based Methods</i>					
TIRG [26] (CVPR'19)	14.61	48.37	64.08	90.03	54.27
CIRPLANT [16] (ICCV'19)	19.55	52.55	68.39	92.38	58.22
ARTEMIS [6] (ICLR'22)	16.96	46.10	61.31	87.73	53.03
<i>VLP Model-Based Methods</i>					
CLIP4CIR [2] (TOMM'23)	38.53	69.98	81.86	95.93	71.58
TGCIR [30] (ACM MM'23)	45.25	78.29	87.16	97.30	77.00
SADN [27] (ACM MM'24)	44.27	78.10	87.71	97.89	76.99
DQU-CIR [28] (SIGIR'24)	46.22	78.17	87.64	97.81	77.46
CaLa [11] (SIGIR'24)	49.11	81.21	89.59	98.00	79.48
SSN [35] (AAAI'24)	43.91	77.25	86.48	97.45	76.27
CASE [14] (AAAI'24)	49.35	80.02	88.75	97.47	78.90
CoVRBLIP [25] (AAAI'24)	49.69	78.60	86.77	94.31	77.34
SPRC [1] (ICLR'24)	51.96	82.12	89.74	97.18	80.25
ENCODER [15] (AAAI'25)	46.10	77.98	87.16	97.64	77.22
CIRLVLM [22] (AAAI'25)	<u>53.64</u>	83.76	90.60	97.93	81.48
CCIN [23] (CVPR'25)	53.41	<u>84.05</u>	<u>91.17</u>	<u>98.00</u>	<u>81.66</u>
MCoT-MVS _(Ours)	55.33	84.75	91.45	98.55	82.52

4 Experiments

4.1 Dataset and Experiment Settings

Dataset. To verify the effectiveness, we conduct extensive experiments on two standard composed image retrieval benchmarks, i.e. CIRR [16] and FashionIQ [31]. CIRR is an open-world scene image retrieval dataset with 36,554 triplets from 21,552 NLVR2 images [20], split 8:1:1 for training, validation, and testing. Fashion-IQ contains 30,134 triplets from 77,684 fashion images, divided into Dress, Shirt, and Tootie categories. Fashion-IQ targets a specialised domain, whereas CIRR emphasises diverse object interactions in real-world scenes. Evaluations on both datasets better reflect the method's real-world generalization ability.

Evaluation metrics. For performance comparison, following prior works [23], we report average recall at rank K (Recall@K). For CIRR, we provide Recall@1, 5, 10, and 50, along with their mean. For FashionIQ, we report Recall@10 and 50 across the three categories and their mean to evaluate the model's generalization within specific fashion domains.

Implementation Details. We use a Qwen2.5-VL-32B-Instruct¹ with a chain-of-thought (CoT) prompting strategy to obtain reasoning content given the multimodal composed query, including the retained text, the deleted text and the potential target texts. We use concurrent MLLM CoT batch inference acceleration, with a parallel rate of more than 1 image per second. For visual feature extraction, we employ the pre-trained CLIP model. We employ Grounded SAM [19] to segment the objects in reference images with the box prediction threshold set to 0.4 and the text matching threshold set to 0.3. During inference, MCoT reasoning takes ~0.85s per query by API, segmentation takes ~0.3s with ~5.5GB GPU memory, and retrieval (with PVRs, IVRS, and WHC) takes ~0.11s with ~6.4GB memory.

¹Qwen2.5-VL-32B-Instruct: <https://huggingface.co/spaces/Qwen/Qwen2.5-VL-32B-Instruct>

Table 2: Performance comparison on FashionIQ from fashion products. The best and the second-best scores are marked in bold and underlined fonts, respectively.

Method	Dresses		Shirts		Tops&Tees		Avg	
	R@10	R@50	R@10	R@50	R@10	R@50	R@10	R@50
<i>Traditional Model-Based Methods</i>								
TIRG [26] (CVPR'19)	14.87	34.66	18.26	37.89	19.08	39.62	17.40	37.39
CIRPLANT [16] (ICCV'19)	17.45	40.41	17.53	38.81	21.64	45.38	18.87	41.53
ARTEMIS [6] (ICLR'22)	27.16	52.40	21.78	43.64	29.20	54.83	26.05	50.29
<i>VLP Model-Based Methods</i>								
CLIP4CIR [2] (TOMM'23)	33.81	59.40	39.99	60.45	41.41	65.37	38.40	61.74
TGCIR [30] (ACM MM'23)	45.22	69.66	52.60	72.52	56.14	77.10	51.32	73.09
FashionERN [5] (AAAI'24)	50.32	71.29	50.15	70.36	56.40	77.21	52.29	72.95
SADN [27] (ACM MM'24)	40.01	65.10	43.67	66.05	48.04	70.93	43.91	67.36
DQU-CIR [28] (SIGIR'24)	<u>57.63</u>	<u>78.56</u>	<u>62.14</u>	<u>80.38</u>	<u>66.15</u>	<u>85.73</u>	<u>61.97</u>	<u>81.56</u>
CaLa [11] (SIGIR'24)	42.38	66.08	46.76	68.16	50.93	73.42	46.69	69.22
SSN [35] (AAAI'24)	34.36	60.78	38.13	61.83	44.26	69.05	38.92	63.89
CASE [14] (AAAI'24)	47.44	69.36	48.48	70.23	50.18	72.24	48.70	70.61
CoVRBLIP [25] (AAAI'24)	44.55	69.03	48.43	67.42	52.60	74.31	48.53	70.25
SPRC [1] (ICLR'24)	49.18	72.43	55.64	73.89	59.35	78.58	54.72	74.97
ENCODER [15] (AAAI'25)	51.51	76.95	54.86	74.93	62.01	80.88	56.13	77.59
CIRLVLM [22] (AAAI'25)	50.42	73.57	58.59	75.86	59.61	78.99	56.21	76.14
CCIN [23] (CVPR'25)	49.38	72.58	55.93	74.14	57.93	77.56	54.41	74.76
MCoT-MVS (Ours)	58.45	78.92	63.24	81.15	68.02	85.97	63.24	82.01

For training on the CIRR dataset, all parameters are fine-tuned with a learning rate of 1×10^{-6} . In contrast, for the FashionIQ dataset, the CLIP parameters are optimized at a learning rate of 1×10^{-6} , while the remaining parameters are trained with a learning rate of 1×10^{-5} to facilitate better convergence. Similar to DQU-CIR [28], we also write reasoned target text in the reference image to improve small modifications of simple images with good performance improvement on FashionIQ verified by DQU-CIR. However, in complex CIRR scenes, the improvement was not significant (only 0.4% in averaged Recalls) after verification. We adopt a fixed batch size of $B = 16$ for all experiments. The temperature factor τ in Eqn. (1) is set to 0.01 for CIRR and 0.1 for FashionIQ. All experiments are implemented using PyTorch and conducted on a single NVIDIA A800 GPU. To ensure reproducibility, the random seed is set to 124 throughout all experiments.

4.2 State-of-the-art Comparisons

We perform extensive experiments to compare our MCoT-MVS with mainstream CIR studies and the latest state-of-the-art methods on two widely-used CIR benchmarks, i.e. CIRR and FashionIQ, in Table 1 and Table 2, respectively.

Quantitative Comparison on CIRR: On the CIRR dataset, which features complex textual modifications and semantically rich images, our MCoT-MVS consistently surpasses mainstream approaches with impressive margins across all evaluation metrics in Table 1. Compared to the latest state-of-the-art CCIN [23], which also employs a pre-trained LLM to reason about user intent by reasoning image-caption-derived queries into conflict and preserved sets, MCoT-MVS achieves a 1.92% gain in Recall@1 and further improves performance at higher cutoffs (e.g., Recall@5 and Recall@50), leading to a 0.86% average improvement across all metrics. These

consistent gains demonstrate that explicitly decomposing and reasoning about user intent allows our MCoT-MVS to better capture multi-level visual references guided by the textual modification, thereby delivering more accurate and robust retrieval results on the challenging benchmark.

Quantitative Comparison on FashionIQ: Table 2 also lists the result of competing methods on the FashionIQ dataset. Our MCoT-MVS also achieves best retrieval performance on all eight evaluation metrics in three categories, including “Dresses”, “Shirts”, “Tops&Tees”, and corresponding average scores. In particular, compared to the latest method CIR-LVLM [22], which fine-tunes an MLLM to refine user intent instructions, MCoT-MVS yields notable gains of 8.1%, 4.6%, and 8.4% in Recall@10 for the “Dresses”, “Shirts”, and “Tops&Tees” categories, respectively. Additionally, compared with the state-of-the-art DQU-CIR under the same data processing settings, our MCoT-MVS achieves improvements of 1.2% and 0.4% in average Recall@10 and Recall@50, respectively.

Overall, these results on two different-domain datasets demonstrate the powerful effectiveness of the proposed MCoT-MVS model with strong generalization capability.

4.3 Ablation Studies

We perform detailed ablation studies on the CIRR dataset to investigate the effectiveness of each component of our proposed MCoT-MVS. The detailed results are shown in Table 3.

The Effectiveness of the Weighted Hierarchical Combination Module. We conduct different feature combinations to observe the performance of different multimodal query feature combinations and evaluate the superiority of our proposed Weighted Hierarchical Combination (WHC) module. First, comparing Baseline 7 and Baseline 8 reveals that using a learnable combiner to fuse multimodal query features leads to notable performance gains. It

Table 3: Ablation study for MCoT-MVS on CIRR. “M-T” and “T-T” indicate the modification texts and the reasoned target texts.

NO.	Variant							Performance				
	WHC	Sum	Combiner	M-T	T-T	PVRS	IVRS	Recall@1	Recall@5	Recall@10	Recall@50	Mean
1	✓	-	-	✓	✓	✓	✓	55.33	84.75	91.45	98.55	82.52
2	-	✓	-	✓	✓	✓	✓	52.02	83.01	90.80	98.36	81.05
3	-	-	✓	-	✓	✓	✓	45.00	76.48	85.35	96.87	75.93
4	-	-	✓	✓	-	✓	✓	54.50	83.69	91.28	98.56	81.98
5	-	-	✓	✓	-	✓	-	53.49	84.07	91.21	98.55	81.83
6	-	-	✓	✓	-	-	✓	53.91	83.81	90.87	98.53	81.78
7	-	-	✓	✓	-	-	-	51.66	82.61	90.21	98.60	80.77
8	-	✓	-	✓	-	-	-	49.61	81.61	89.74	98.55	79.87

suggests that composed queries inherently contain modality intent ambiguity, which benefits from adaptive weighting. Second, by comparing full MCoT-MVS (No.1) with Variant NO.2, where the proposed WHC module is replaced by direct summation of four types of multimodal query representations, the retrieval performance substantially drops from 82.52% to 81.05% in average recall. These observations demonstrate that our WHC module not only effectively integrates the intent information from the modified text (e.g. appearance of new objects) but also effectively filters out potential errors from target reasoning, while preserving the effective selection of the reference image.

Effects of the MLLM-based Reasoned Target Text. To evaluate the impact of potential target content reasoned from the multimodal composed query via MLLM, we first directly replace modification texts with the reasoned target texts to retrieve target images (indicated Variant No.3). This variant achieves an average retrieval performance of 75.93%, demonstrating the validity of the reasoned content. However, its effectiveness is still lower than that of using the original modification text, which we attribute to the fact that multimodal reasoning may introduce a certain degree of hallucination, while modification text generally provides more precise intent cues. Nevertheless, the results confirm that the reasoning ability is meaningful and beneficial. By comparing the performance of our MCoT-MVS (No.1) with Variant 3 and Variant 4, it is evident that integrating the reasoned target content alongside the original modification text provides additional accurate and effective intent information for alignment with the target image. It also proves the effectiveness of our multimodal CoT reasoning.

The Effectiveness of Multi-level Visual Reference Selection (PVRS and IVRS). We construct extensive ablation studies, including experimental variants No.4 to No.7, to evaluate the impact of our proposed patch-level visual reference selection module (PVRS) and instance-level visual reference selection module. Specifically, when removing the PVRS module, the Recall@1 performance decreases absolutely by 0.59% compared with variant No.4. It suggests that patch-level reference image de-noising guided by the reasoned retained and removed user intent plays a vital role in our model. Similarly, when removing the IVRS module, it leads to an absolute 1.01% drop in Recall@1. By independently modelling instances in the reference image, the key instances of user intent can be clarified sufficiently and effectively, and the user intent representation can be enhanced from the overall semantics and even the spatial shape of the reference image instances. Furthermore, when both PVRS and IVRS are removed, the overall retrieval performance drops substantially from 81.98% to 80.77%, demonstrating that the two

Table 4: Retrieval performance (%) of MCoT-MVS with different reasoning patterns.

Model	R@1	R@5	R@10	R@50
w/o Multi-step MCot	54.29	84.45	91.15	98.60
w/ Multi-step MCot	55.33	84.75	91.45	98.55

Table 5: Retrieval performance (%) of MCoT-MVS with different reasoning models.

Model	R@1	R@5	R@10	R@50
Llama-3.2-11B ¹	54.31	84.26	91.78	98.69
Qwen2.5-VL-7B ²	54.53	84.31	91.13	98.50
Qwen2.5-VL-32B	55.33	84.75	91.45	98.55

modules are complementary. These results strongly validate the effectiveness of our proposed multi-level visual reference selection mechanism in MCoT-MVS for improving the interpretation and clarification of multimodal composed queries in the CIR task.

4.4 The Impact of Chain-of-Thought Reasoning

We conduct the comparison experiments in Table 4 to validate the importance of multi-step multimodal chain-of-thought (MCoT) reasoning. In particular, when reasoning about the explicit use intent without the multi-step CoT, i.e., directly letting the MLLM predict the “retain/delete/target” pattern, the retrieval performance decreases, especially on the higher top-ranking. The main reason for this is that multi-step chain-of-thought reasoning can yield a more precise decomposition of user intent. Although performance without a multi-step MCoT is not optimal, the displayed user intent decomposition still outperforms the baseline (No. 7 in Table 3) without it. This further demonstrates the effectiveness of displaying the decomposed user intent and making reference visual selections.

4.5 The Impact of Reasoning Abilities

To further examine the sensitivity of our proposed MCoT-MVS framework to the reasoning abilities of different MLLMs, we conduct experiments under three representative multimodal reasoning models: Llama-3.2-11B-Vision-Instruct, Qwen2.5-VL-7B, and Qwen2.5-VL-32B.

As shown in Table 5, MCoT-MVS consistently achieves strong retrieval performance across all reasoning models. Although the larger Qwen2.5-VL-32B achieves the best overall accuracy (R@1:

¹Llama-3.2-11B: <https://huggingface.co/meta-llama/Llama-3.2-11B-Vision>

²Qwen2.5-VL-7B: <https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>

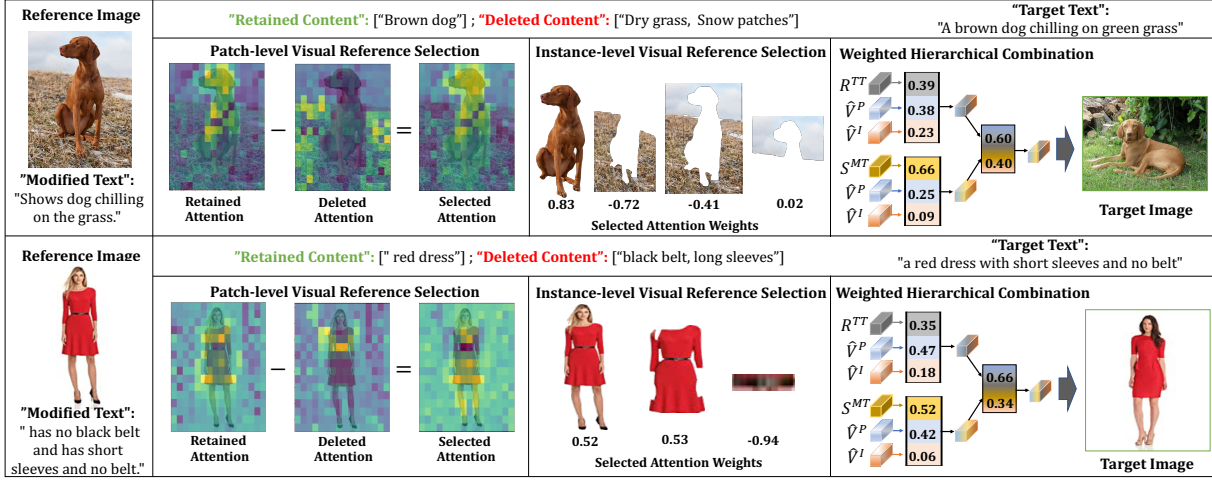


Figure 3: Visualization of the learned attention weights from PVRS and IVRS modules guided by the inferred explicit retain or delete modification intents from the MLLM-based multimodal CoT reasoning, as well as the weight distributions of weighted hierarchical combination modules (best viewed in color).

55.325), the smaller Llama-3.2-11B and Qwen2.5-VL-7B still maintain highly competitive results. This demonstrates two key findings: (i) MCoT-MVS has a certain sensitivity to improvements in reasoning power, as more powerful reasoning models can lead to retrieval improvements by better understanding user intent; and (ii) MCoT-MVS is robust to weaker reasoning ability, as performance does not collapse even when reasoning power decreases, maintaining strong performance compared to other models. These results demonstrate that MCoT-MVS effectively constrains reasoning output and maintains reliability across varying model capacities.

4.6 Qualitative Analysis

To better understand the contribution of each component in MCoT-MVS, we visualize the attention weights from PVRS and IVRS modules in Figure 3, guided by multimodal CoT reasoning. Specifically, the PVRS module performs semantic denoising at the patch-level by enhancing areas aligned with the preserved intent and attenuating those related to modified or deleted content, thus enabling more fine-grained visual filtering. Meanwhile, the IVRS module operates at the high-level instance-level to capture key target instances mentioned in the user's intent by modeling their appearance features while discarding distracting instances. These attention maps clearly illustrate the complementary effects of the two modules at different granularity levels, leading to enhanced visual intent representation and ultimately improved query interpretation and retrieval performance in CIR tasks.

Additionally, we also visualize the learned weights from WHC module. In both examples, the reasoned target text receives higher weights than the original modification text, better capturing user intent and validating the effectiveness of our multimodal CoT reasoning. For visual references, the patch-level selected features are assigned greater weights due to their finer granularity, while the instance-level features also provide meaningful complementary cues. These results demonstrate that WHC can effectively allocate fusion weights across multimodal inputs, enhancing intent representation and improving CIR performance.

5 Conclusion and Future Work

In this paper, we propose MCoT-MVS, a novel method for Composed Image Retrieval (CIR) that explicitly selects multi-level reference visual features under the guidance of multimodal Chain-of-Thought (CoT) reasoning. By disentangling and reconstructing fine-grained and high-level visual cues based on explicit retain or delete modification intents, our approach significantly enhances visual intent representation. Furthermore, a weighted hierarchical attention combination mechanism (WHC) effectively integrates the modified text, reasoned target content, and multi-level visual selected references, enabling the model to suppress irrelevant information and preserve a coherent fusion aligned with the target image. Extensive quantitative evaluations on two standard CIR benchmarks, CIR and FashionIQ, demonstrate that MCoT-MVS achieves state-of-the-art performance across multiple retrieval metrics.

On one hand, while the target content of multimodal CoT reasoning is available, its effectiveness still needs to be improved. On the other hand, multi-step MCoT reasoning has a certain impact on reasoning efficiency, and direct reasoning also provides good performance for user intent. Therefore, in future work, we will further explore hybrid reasoning paradigms to improve the accuracy of target reasoning while further enhancing the efficiency of the reasoning process.

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